A Hybrid Deep Learning Approach for Signature Verification

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***Abstract*—Offline handwritten signature forgery remains a challenging problem in document image forensics because of the high similarity in appearance between genuine and forged samples. While financial, legal, and identity-based applications require verification to be reliable, traditional handcrafted methods often cannot generalize well across diverse writing styles. This work proposes a hybrid deep learning framework that fuses EfficientNet-based CNN embeddings, PCA-compressed HOG, and LBP features under a Siamese network trained with triplet loss. The proposed model yields a strong verification accuracy on dataset, outperforming handcrafted-only methods by effectively modeling global stroke patterns and fine-grained texture cues. These results indicate that hybrid embedding methods are suitable for secure and efficient offline signature authentication.**

***Index Terms*—Signature verification, Siamese networks, deep learning, handcrafted features, metric learning, PCA, forgery detection, biometric authentication**

I. Introduction

Although many legal, banking, and governmental processes are becoming digital, a handwritten signature remains a requirement for authenticating documents. Improvements in image processing and reproduction techniques have made it relatively easy to create signature forgeries that bear a high degree of credibility, which is a serious security concern for identity verification and the authenticity of official documents. Signature verification techniques that rely on geometric or texture-based features created by the human hand have inherent weaknesses that apply to a broader range of exemplars are vulnerable to expert forgeries and subject to modifications or distortions resulting from the scanning process or acquisition conditions [1]. As shown in Fig. 1, genuine and forged signatures can appear visually similar, making manual verification highly challenging.

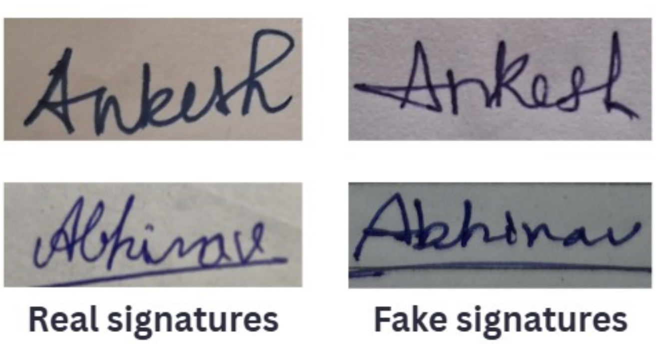

Fig. 1. Demonstration real and fake images.

A. Motivation for Using Deep Learning

Deep learning, especially convolutional neural networks embedding metric learning, is a more robust alternative because it learns discriminative, hierarchical representations directly from signature images [2]. There are hybrid models that combine consequences of deep embeddings with handcrafted descriptors that have been shown to enhance robustness by capturing global stroke characteristics and local texture information together [3]. Therefore, this project proposes a hybrid Siamese-based model proposed to integrate CNN features with compressed handcrafted representations to improve the robustness of systems that perform offline signature verification.

B. Objectives of the Project

The goal of this project is to use a hybrid deep learning framework to create an efficient offline signature verification system. The particular goals are:

- To preprocess and standardize signature datasets using conversion, cleaning, and normalization procedures to guarantee format quality and consistency.
- To extract and compress handcrafted features using HOG, LBP, and Incremental PCA for efficient representation [4].
- To use triplet-loss metric learning to create and train a hybrid Siamese architecture that combines handcrafted descriptors with CNN-based embeddings [5].
- To use important metrics like accuracy, ROC-AUC, threshold-based classification, and embedding distance analysis to assess verification performance.
- To evaluate the model's suitability for practical document authentication applications and how well it detects skilled forgeries.

The rest of the paper is structured as follows. Section 2 summarizes the related work on offline signature verification using handcrafted feature-based methods, classical machine learning, and recent deep metric-learning approaches. Section 3 describes the proposed methodology, which deals with dataset preparation, handcrafted feature extraction, hybrid Siamese architecture design, and the training evaluation pipeline. Section 4 covers experimental results including ROC analysis, distance distributions, validation metrics, and generalization to unseen writers. The paper's main conclusions, limitations, and possible future directions to enhance the performance of hybrid signature verification systems are finally covered in Section 5

II. Related Work

Traditionally, offline signature verification systems used handcrafted geometric or texture-based features such as contour descriptors, stroke width measures, HOG, and LBP. Although these achieved reasonable performance, they depended on manual feature engineering, and were susceptible to skilled forgeries, noise, and other variations related to natural writing, which did not allow for subtle variations intra-writer.

To enhance differentiation of authentic samples and counterfeit samples, researchers explored basic machine learning algorithms such as support vector machines (SVM), hidden Markov models (HMMs), and ensembles [6]. While these algorithms produce more viable generalizations than handcrafted inductive rule-based systems, they were inherently inhibited by the capacity of the fixed feature extractors, and are limited to their ability to represent complex stroke structure and texture variance.

Deep learning produced stronger methods through the use of Siamese networks, which learn distance-based similarity functions instead of explicit class labels, and also allow for writer-independent verification. Later developments showed triplet-loss metric learning, like FaceNet, could provide stronger separation between genuine and skilled forgeries in the embedding space.

Studies conducted more recently examined hybrid methods that integrate CNN embeddings with handcrafted descriptor embeddings, which increase robustness to variability in writing style and differences in image quality. [3]. Writer-independent techniques continue to report strong results with deep neural networks while promoting efforts to create more discriminative and robust embedding models better suited for real-word signature variability. [5]

III. Methodology

A. Dataset Preparation, Handcrafted Features, and Preprocessing

For this project, a completely unique offline signature dataset was created to guarantee writer diversity and authentic forgery variations. Ten authentic signatures and ten fake signatures were obtained for each of the 173 participants. Another participant attempted to replicate the authentic writing style as closely as possible in order to create the forged signatures. This produced 3,460 signature samples overall, with 20 signatures per writer.

The raw images were initially accumulated in a mixed format (JPG, PNG, HEIC, TIFF, and scanned PDFs). To standardize the dataset, each signature sample was transformed into JPEG format and any images that were holographic, cropped, broken, or blurred were manually deleted. All images were resized to 224 x 224 pixels to be compatible with CNN models.

The data were then sorted into writer-specific folders, each containing separate directories for real and forged signatures. To enhance generalization to the variability of capturing handwriting in the wild, we applied a series of data augmentation techniques namely, random rotations (up to ±360 degrees), affine transformations, brightness jitters, and random erase, for example, to each image in the datasets. This systematic processing produced a dataset that was all standardized, high-quality, and well-suited to Siamese based metric learning.

TABLE I
Energy-Aware Scheduling Dataset Statistics

Statistic	**Value**
Total Writers	173
Genuine Signatures per Writer	10
Forged Signatures per Writer	10
Genuine Signatures	1,730
Forged Signatures	1,730
Total Signatures	3,460
Image Format	JPEG (.jpg)
Standardized Resolution	224×224 pixels
Category Labels	Real / Forged

After the preparation of the dataset, handcrafted feature descriptors were calculated for each signature image. Histograms of Oriented Gradients (HOG) and Local Binary Patterns (LBP) were employed to capture contour structure, stroke directionalities, and texture differences that may be representative of acceptable model-based forgery expertise, to a degree. The descriptors were joined together to build high dimension feature vector. In turn, to conserve memory and reduce redundant artifacts, we opted to complete Incremental Principal Component Analysis (IPCA) in streaming mode, so we could preserve critical discriminative feature structures represented in a compact 64 dimension representation of the handcrafted features.

Simultaneously, all signature images underwent a data augmentation process involving random rotations, affine transformations, color jittering, and erasing. This was done to train the model to be more robust to noise and small distortions introduced during the scan/acquisition process, as well to simulate more natural variability in handwriting. Each augment section was then normed and prepared as tensors for use with deep convolutional neural networks.

B. Hybrid Siamese Model, Training Strategy, and Evaluation

The suggested hybrid Siamese model is intended to produce discriminative embeddings through the combination of deep convolutional features and compact handcrafted descriptors. Each branch of the Siamese model has two pathways that are parallel to each other: one utilizing a deep feature extractor which is based on EfficientNet-B0, designed to capture high level structures and stroke patterns; and the other featuring handcrafted representations where HOG–LBP features are first reduced to 64 dimensions through Incremental PCA, before projected to an embedding space of 128 dimensions. The deep and handcrafted representations are fused, and delivered through a final linear layer to produce a 128-dimensional signature embedding, normalized across all channels.

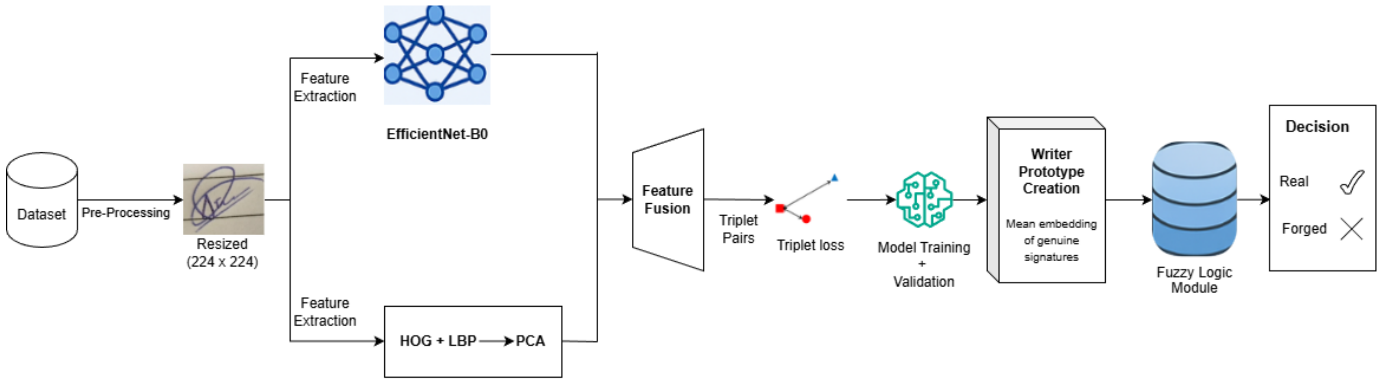

Fig. 2. Overall Hybrid Signature Verification Pipeline including pre processing, feature extraction, triplet-loss training, and prototype-based verification.

Algorithm 1 CNN-Based Feature Extraction using EfficientNet-B0

1: **Input:** Preprocessed image P
2: **Output:** Deep feature embedding vector E_{cnn}
3: Initialize EfficientNet-B0 backbone
4: Feed input image P into the EfficientNet-B0 network
5: Extract intermediate convolutional feature maps
6: Apply Global Average Pooling (GAP) to reduce spatial dimensions
7: Pass pooled feature vector through Dense projection layer
8: Perform L2-normalization on the output vector
9: **return** E_{cnn}

1) CNN-Based Feature Extraction Algorithm:

Algorithm 2 Handcrafted Feature Extraction (HOG + LBP + PCA)

Require: Processed image P
Ensure: Handcrafted feature vector E_{hand}
1: Convert P to grayscale image G
2: Compute HOG features on G
3: Compute LBP features on G
4: Convert LBP to histogram H
5: Concatenate HOG and H into a single vector
6: Optionally apply PCA for dimensionality reduction
7: **return** E_{hand}

2) Handcrafted Feature Extraction Algorithm:

C. Triplet Loss Formulation

The Siamese network used in this work is trained with Triplet Margin Loss, which promotes that the embeddings of real signatures will remain close to one another while the forged actually move farther apart from each other in the embedding space. A triplet consists of an anchor (x_a), a positive sample (x_p) from the same writer, and a negative sample (x_n) which is a forged signature.

Let $f(\cdot)$ denote the embedding function that maps an image to its feature representation. The Triplet Loss is defined as:

$$\mathcal{L}_{triplet} = \max\left(0, \begin{array}{l} \|f(x_a) - f(x_p)\|_2^2 \\ - \|f(x_a) - f(x_n)\|_2^2 + \alpha \end{array}\right) \tag{1}$$

where α is the margin that enforces a minimum distance between the positive and negative pairs.

The objective of the loss is to ensure:

$$\|f(x_a) - f(x_p)\|_2 + \alpha < \|f(x_a) - f(x_n)\|_2 \tag{2}$$

This encourages tight clustering of genuine signatures while simultaneously pushing forged signatures farther away. During inference, embeddings are L2-normalized:

$$f(x) \leftarrow \frac{f(x)}{\|f(x)\|_2} \tag{3}$$

to ensure stable, scale-invariant distance comparisons.

Algorithm 3 Triplet Loss Training Algorithm

Require: Triplets (A, P, N) consisting of Anchor, Positive, Negative
Ensure: Updated network parameters
1: **for** each batch of triplets **do**
2: Compute embeddings $E_A = model(A)$
3: Compute embeddings $E_P = model(P)$
4: Compute embeddings $E_N = model(N)$
5: Compute $d_{pos} = distance(E_A, E_P)$
6: Compute $d_{neg} = distance(E_A, E_N)$
7: Compute loss

$$loss = \max(d_{pos} - d_{neg} + margin,\ 0)$$

8: Backpropagate loss and update model weights
9: **end for**

1) Triplet Loss Training Algorithm:

D. Training and Evaluation

To guarantee that no author was included in more than one subset, the dataset was subset after a writer-disjoint split and it was divided into training, validation, testing sets as well. The hybrid Siamese model was trained with the AdamW optimizer with a learning-rate of 3e-4 and with Triplet Margin Loss to enforce similarity to real signatures while separating from forged signatures.

During evaluation, writer prototypes were formed by averaging embeddings of genuine signatures, and verification was performed by measuring the distance between each test embedding and its corresponding prototype. Performance was evaluated using accuracy, ROC–AUC, and optimal threshold selection based on distance distributions, enabling reliable discrimination between genuine and forged signatures.

IV. Discussion of Results

The diagnostic plots illustrate the proposed hybrid Siamese model, and they reveal if the model is capable of being used to detect offline signature forgery. The model is based on distance-based metric learning, in which signatures that come from the same writer should group together closely in the embedding space while forged signatures should cluster apart in the embedding space. The figures and metrics provided in this section show how efficient the proposed hybrid Siamese model was able to accomplish this task.

A. Validation ROC Analysis

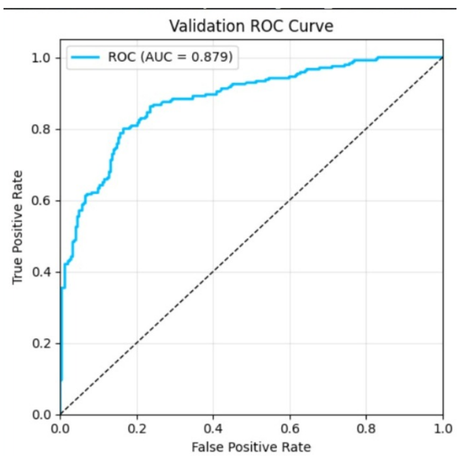

Fig. 3. Validation ROC Curve showing the trade-off between True Positive Rate and False Positive Rate.

Figure 3 shows the ROC curve of the validation set. In signature verification, the ROC curve provides an evaluation that is independent of threshold, by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR). At the point of the ROC curve, the model achieves an AUC of 0.879, indicating the discriminative ability of the model is very strong. The large increase in the curve towards the upper-left corner indicates that over a wide range of thresholds, the model is successful at correctly identifying genuine signatures and simultaneously not accepting forgeries as genuine. The diagonal reference line represents identification by chance, and the clear distance above this line demonstrates that the learned embedding space has captured meaningful identity-specific features.

B. Validation Metrics Summary

TABLE II
Validation Evaluation Metrics

Metric	**Value**
Accuracy	81.54%
AUC	0.879
Optimal Threshold	0.709
Standard Deviation (σ)	0.273

Table II summarizes the validation performance. The optimal threshold of 0.709 is determined from the analysis of distance distributions as the point at which the classification becomes maximally correct. In distance-based forgery detection, this threshold indicates the boundary between “genuine” and “forged” signatures, and this particular threshold value is well supported by the data. Because the standard deviation of distances is relatively low, it implies a stable level of intra-writer similarity which argues that the embedding space retains compact clusters of genuine samples.

C. Validation Distance Distribution

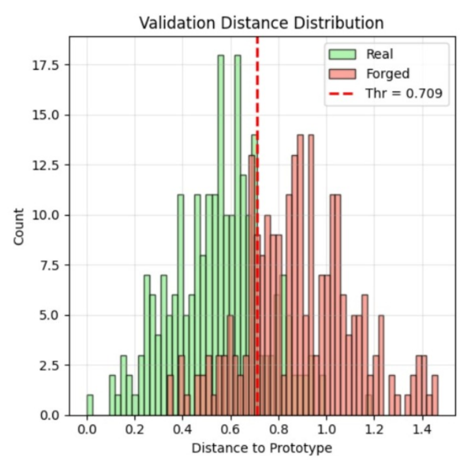

Fig. 4. Validation Distance Distribution for Real and Forged Signatures.

Figure 4 shows the difference between authentic and forged signatures according to their distance to the writer prototype. In metric-learning systems, smaller distances are indicative of stronger similarity. As expected, authentic signatures cluster to the left of the threshold and forgeries are to the right. While some overlap exists - an indication of the difficulty of highly skilled forgeries – the general distribution of signatures shows the triplet-loss is achieving what it is intended to do, create small intra-class distances and large inter-class distances.

D. Test ROC and Distance Analysis

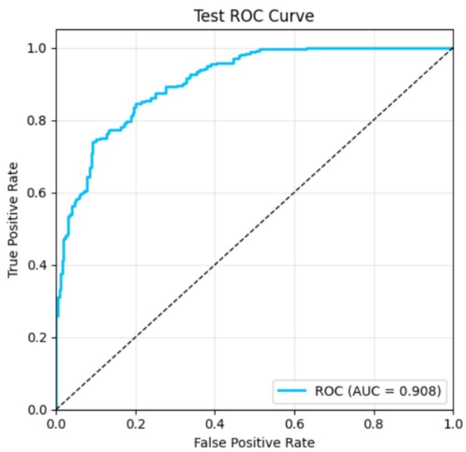

Fig. 5. Test ROC Curve indicating strong generalization performance.

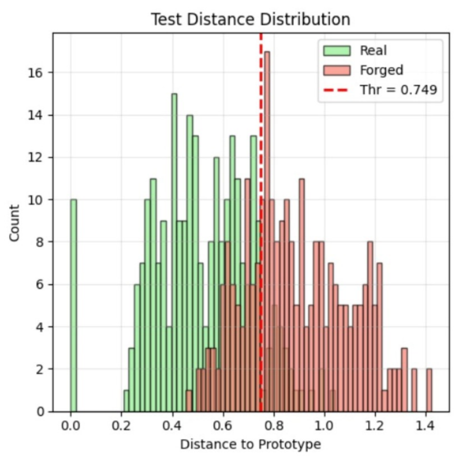

Fig. 6. Test Distance Distribution with Optimal Threshold.

Figures 5 and 6 show the model's accuracy on out-of-sample authors. The area under the curve (AUC) of the ROC for the test set is 0.908, meaning that the model generalizes reasonably well from the original authors to the test sample of other authors. Generalization/generalizability is a necessary attribute of a well-functioning verification system, and with a case in front of it, it must be able to verify signatures from characters/ authors it has never seen before. The distribution of test distances was analogous to the validation, including clear clusters of valid signatures and a wider dispersion of forged signatures.

E. Overall Interpretation

The theoretical basis of the model rests on the ability to distinguish distance:

- Triplet Loss encourages real signatures to get closer together in embedding space while moving forged signatures further apart.
- Prototype Matching utilizes distances computed class-wise by centroids, a typical practice in metric learning to decrease writer variability.
- Threshold-Based Verification applies a learned distance cutoff to classify signatures without retraining for new writers.

This principle holds true in both the validation and test results. Authentic signatures always yield low prototype distances, while forged signatures yield significantly higher ones. The ROC analysis shows a strong discriminative ability, and the accuracy based on the threshold indicates that the system performs well and provides consistent results. Together, these results show that the hybrid Siamese system learns and captures the necessary for offline signature verification, both structurally and texturally.

V. Conclusion

We developed a hybrid Siamese verification model that fuses handcrafted descriptors with deep convolutional features using a writer-disjoint dataset of genuine and forged signatures. The model shows strong discriminative ability, achieving high AUC scores and clearly separating genuine and forged signatures in the embedding space. The ROC and distance analyses further confirm that the learned representations generalize well to unseen writers, enabling reliable threshold-based verification. Overall, the proposed hybrid architecture provides a practical and effective solution for offline signature authentication, improving security and consistency in administrative, financial, and legal applications.

References

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